

Agentic Workflow — Complete Reference

Last updated: 2026-08-16 · Living document, edit this directly for day-to-day changes. As of 2026-08-16 this reference is version-controlled at `docs/agentic-workflow.md` inside the `dotfiles` repo (it previously sat loose in the untracked parent folder), so it clones to a fresh machine with the rest of the setup. `agentic-workflow.pdf` is a periodic polished snapshot generated from this file's content; only worth regenerating after a batch of real changes, not every edit. Regenerate it with `docs/build-pdf.sh` (Markdown -> HTML -> headless-Chrome print-to-PDF).

Current state: fully built and verified working. WezTerm + herdr + Neovim + Claude Code + Codex + OpenCode, all wired to local (`omlx`) and cloud (Claude) inference, voice input via Parakeet v3, the full axi tool family, and Kun's productivity stack (lavish, treehouse, no-mistakes, gnhf, firstmate) are all installed, declared in Nix, and tested end-to-end. GitHub Copilot CLI (`gc`) was added 2026-07-11 and authenticated 2026-08-16, now verified working end-to-end, see §2. Codex's default local model swapped again 2026-07-12, to `mlx-community--Qwen3-Coder-Next-8bit` — see §5. That session also uncovered and fixed a real bug in `rebuild.sh` (missing `--impure`) that had been silently zeroing out `OMLX_API_KEY` on every rebuild — see §9.

1. Quick Reference — Keys & Commands

The one mental model: a prefix key (herdr = `Ctrl+b`) precedes multiplexer commands. In Neovim, a leader key (`Space`) precedes custom commands. Press prefix/leader, release, then the next key.

herdr — prefix `Ctrl+b`

Key	Action
<code>Ctrl+b "</code>	split pane horizontal
<code>Ctrl+b %</code>	split pane vertical
<code>Ctrl+b h/j/k/l</code>	move focus
<code>Ctrl+b c</code>	new tab (fresh shell)
<code>Ctrl+b &</code>	close tab
<code>Ctrl+b w</code>	workspace picker
<code>Ctrl+b g</code>	go to (jump)
<code>Ctrl+b y</code>	copy mode (v/Space select, y/Enter copy)

Sessions persist on a background herdr server — closing the terminal doesn't kill your work. But see §9 for why that same persistence can cause stale-environment bugs.

Neovim — leader `Space`

Key	Action
<code>h j k l</code>	left/down/up/right
<code>w / b</code>	next/prev word
<code>0 / \$</code>	start/end of line
<code>gg / G</code>	top/bottom of file
<code>Nj / Nk</code>	jump N lines (relative #s in gutter)
<code>i / a / o / O</code>	insert modes
<code>Esc</code>	leave insert + saves (custom map)
<code>dd / yy / p</code>	delete / copy / paste line
<code>u / Ctrl+r</code>	undo / redo

Key	Action
Space f	find files
Space s	search/grep project
Space b	switch buffers
Space e	file browser (Oil)
Space g	git UI (Neogit)
Space m	markdown preview, floating window (glow.nvim), :q to close
gd	go to definition

Agents

Alias	Command
cc	claude --dangerously-skip-permissions (cloud)
co	codex --sandbox workspace-write --ask-for-approval never (routed to local omlx, default model mlx-community--Qwen3-Coder-Next-8bit as of 2026-07-12 — see §5)
oc	omlx launch opencode --model mlx-community--Qwen3-Coder-Next-8bit (routed to local omlx). Swapped off Qwen3.6-35B-A3B on 2026-08-17 to match co, see §5. Prefer this over co for context-heavy work.
ou	omlx launch opencode --model Qwen3.8-27B-Uncensored-8bit — deep reasoning + refusal bypass, ~18 tok/s, see §5a
gc	copilot --allow-all (cloud, GitHub Copilot subscription). Added 2026-07-11, authenticated 2026-08-16 and working, see §2.

/ — slash commands (skills, config, /lavish, /no-mistakes). Plan mode — ask the agent to plan before building. Esc — interrupt agent mid-task.

Voice input (Parakeet v3)

- hold `⌘ Option Right` — push-to-talk: hold, speak, release
- `⌘ Option + `` — toggle: press to start, press to stop

Global hotkey — works in any focused app (herdr pane, Messages, anywhere), agent-agnostic. No vocab-priming with Parakeet (Whisper-only feature) — see §6.

Git essentials

Command	Action
git status	what changed
add	alias: git add .
git commit -m	commit staged
push / pull	aliases for git push/pull
m	alias: git switch main
lazygit	full-screen git TUI

Protocol: HTTPS via `gh auth login`, not SSH — background daemons (no-mistakes, gnhf, firstmate crew) can't reliably reach an SSH agent socket, but the gh-issued token in Keychain works for any process.

omlx (local inference)

Command	Action
omlx status	server/client health

Command	Action
<code>omlx restart</code>	restart + rescan for new models
<code>omlx launch <tool> --model <id></code>	wire a tool to omlx
<code>hf download <repo></code>	pull a new model (omlx's bundled CLI, <code>/opt/homebrew/Cellar/omlx/<version>/libexec/bin/hf</code>)

2. Component Status

Component	Status	Notes
WezTerm, herdr, Neovim, Claude Code, node/npx	done	core ship, unchanged since original setup
lavish	done	project + global skill; visual planning
skill-creator	done	Anthropic's own — lets agent author new skills
gh-axi	done	GitHub ops, ~3x cheaper/2x faster than GitHub MCP
chrome-devtools-axi	done	browser automation, ~40% fewer tokens than raw MCP
quota-axi	done	usage tracking across Claude/Codex/Cursor/Copilot/Grok
treehouse	done	pooled git worktrees, <code>~/local/bin/treehouse</code>
no-mistakes	done	validation pipeline, daemon running
gnhf	done	"good night, have fun" — the unattended overnight autonomous loop. Via <code>npx gnhf</code> (no global install — see §9)
firstmate	done	cloned to <code>~/github/kunchenguid/firstmate</code> ; fresh-machine re-clone step now documented in the dotfiles README
gh (GitHub CLI)	done	authenticated, HTTPS protocol
omlx	done	v0.5.0. GPU memory ceiling + context-window policy hardened 2026-07-10 — see §5
Codex CLI → omlx	done	wired to omlx, verified via <code>codex exec</code> . Bundled model catalog doesn't recognize local models — cosmetic metadata warning only, see §5
OpenCode (oc) → omlx	done	wired via <code>omlx launch opencode</code> ; preferred over Codex for context-heavy sessions since omlx writes real metadata into its config — see §5
Claude Code → omlx	decided	deliberately cloud-only — not wired, by choice
GitHub Copilot CLI (<code>gc</code>)	done	cask <code>copilot-cli</code> declared in <code>configuration.nix</code> , alias in <code>home.nix</code> , installed via <code>./rebuild.sh</code> 2026-07-11. Authenticated via <code>copilot login</code> and verified working end-to-end 2026-08-16. Cloud-only for now; it does support BYOK to an OpenAI-compatible endpoint via <code>COPILLOT_PROVIDER_BASE_URL</code> (<code>copilot help providers</code>), so wiring it to omlx like <code>co / oc</code> is possible but not yet attempted — see §5. No global instructions file (no <code>~/copilot/AGENTS.md</code> equivalent) — see §3.
Markdown preview (<code>glow</code> + <code>glow.nvim</code> , <code>Space m</code>)	pending	<code>glow</code> added to <code>home.packages</code> in <code>home.nix</code> 2026-07-11; <code>glow.nvim</code> added as new <code>lua/plugins/markdown.lua</code> , lazy-loaded on first <code>:Glow</code> call. Declared, not yet applied - run <code>./rebuild.sh</code> to install <code>glow</code> , then launch <code>nvim</code> once so <code>lazy.nvim</code> fetches the plugin.
leetcode.nvim (<code>:Leet</code>)	done	<code>kawre/leetcode.nvim</code> added as <code>lua/plugins/leetcode.lua</code> 2026-07-11 — no Nix/ <code>rebuild.sh</code> involved, this repo's Neovim plugins are managed entirely by <code>lazy.nvim</code> (installs from GitHub on next <code>nvim</code> launch). <code>lang = 'python3'</code> , <code>plugins.non_standalone = true</code> so <code>:Leet</code> works inside a normal session with other buffers open — close it with <code>:Leet exit</code> , not <code>:q</code> . Sign-in needs the full <code>Cookie</code> request header (<code>csrftoken</code> + <code>LEETCODE_SESSION</code>) copied from DevTools Network tab, not <code>Set-Cookie</code> — see §9 gotchas for the paste/Brave quirks.

Component	Status	Notes
OpenSuperWhisper + Parakeet v3	done	installed, permissions granted, verified pasting text
Kokoro TTS (voice output)	not built	bundled in omlx, unused — see §8

3. Memory Files & Skills — How the Agent Learns Your Preferences

Global memory (`~/.claude/CLAUDE.md` , symlinked to AGENTS.md for Codex/OpenCode too) — personal preferences, loaded into every session. Keep it short (~27 lines); it costs tokens on every request. Yours already has: never use em dash, don't weight development cost in technical decisions, reproduce bugs end-to-end before fixing, be pixel-perfect on UI, fix lint/test issues you notice even if unrelated to the task.

Copilot CLI is the odd one out: as of 2026-07-11, GitHub's docs describe no user-level/global instructions file for `gc` (no `~/.copilot/AGENTS.md` equivalent to symlink `home/AGENTS.md` to). It does read per-repo `.github/copilot-instructions.md` , `.github/instructions/**/*.instructions.md` , and a repo-root `AGENTS.md` (this dotfiles repo already has one) — so it inherits project conventions, just not your personal global preferences. `copilot init` can scaffold the per-repo file. Revisit if GitHub adds a global option later.

Project memory (`CLAUDE.md` / `AGENTS.md` in a repo root) — grows by correcting the agent and telling it to "remember." This is the collective learning of every session in that project.

Skills — conditionally-loaded procedures (only read when relevant), installed via `npm skills add <owner/repo> --skill <name> [-g]` . Move verbose/conditional instructions out of memory files and into skills to keep the memory file lean.

Skill security: a skill can run anything on your machine — API key exfiltration, arbitrary shell access. Do not install skills from unverified sources, even popular ones (a benchmark cited in Kun's video showed a 177k-star skill making agent output measurably worse while using 5% more tokens). Only the official kunchenguid axi catalog (axi.md) and Anthropic's own skills were installed here. Scanner "Med Risk" ratings on gh-axi / chrome-devtools-axi / quota-axi reflect that they can act on GitHub/a browser/your usage data — that's the capability working as intended, not a red flag on the code itself.

4. The Workflow Loop

1. **Plan** — Discuss in lavish (interactive HTML artifact, browser-based annotation) — decide requirements here, not by reading walls of text
2. **Build** — Agent works the middle — step away, start other tasks
3. **Validate** — no-mistakes: branch → adversarial review → e2e test (with evidence) → docs → lint → PR, babysat until merged
4. **Judge** — Review the PR, weighted by its risk assessment — not every diff
5. **Scale** — treehouse (parallel worktrees, no manual cleanup) + firstmate (manages the crew so you talk to one agent, not many)

Captain's mindset: you're an engineering director, not a diff-reviewer. Set good process, trust the pipeline, spend your energy on what to build — competition, users, direction — not line-by-line review.

Daily muscle-memory drills: split a herdr pane, run an agent in one, nvim in another · jump with relative line numbers (`Nj` / `Nk`) instead of scrolling · `Space f` a file, `Space s` a symbol — no mouse, ever · dictate instead of type whenever the prompt isn't a URL/path · detach herdr, reattach — the session is still there.

Goal: hands never leave the keyboard. That's the whole point — staying in flow.

5. Local Inference — omlx

omlx is an MLX inference server purpose-built for coding agents on Apple Silicon: continuous batching, paged SSD KV-caching (repeated large contexts go from 30–90s time-to-first-token down to 1–3s), and native OpenAI + Anthropic API compatibility. Verified legitimate via a real discussion thread in Apple's own ml-explore/mlx GitHub repo.

Current setup

- Installed via private Homebrew tap `jundot/omlx`, trusted and declared in `configuration.nix`
- Running as a background service (`homebrew.mxcl.omlx` launchd job), auto-starts at login
- **Codex's default model as of 2026-07-12:** `mlx-community--Qwen3-Coder-Next-8bit` (80B total / 3B active MoE, 256K native context). Replaced `mlx-community--Qwen3.6-35B-A3B-8bit` — see "Model swap: → Qwen3-Coder-Next-8bit" below for rationale.
- `oc` /OpenCode was swapped to `mlx-community--Qwen3-Coder-Next-8bit` on 2026-08-17, matching `co`. That alias hardcodes its model in `home.nix` rather than reading `~/codex/config.toml`, so it needed a deliberate edit plus a rebuild.
- Codex CLI is wired to it (`~/codex/config.toml`, provider `omlx`, base_url `http://127.0.0.1:8000/v1`) — verified working end-to-end via `codex exec`
- OpenCode is also wired to it via `omlx launch opencode --model <id>` (the `oc` alias) — preferred over Codex for context-heavy work, since omlx writes its real, live context-window metadata into OpenCode's config at launch time instead of Codex guessing from a static bundled catalog
- Claude Code stays cloud-only — a deliberate choice, not a gap.

Trust lesson from this setup: a model already on disk (`...Claude-4.6-Opus-Deckard-Heretic-Uncensored-Thinking-8bit`) was deleted before use. "Heretic" is a known automated ablation technique that strips safety/refusal training — it measurably degrades coherence and instruction-following even when done carefully, and there's no legitimate coding/planning task that benefits from an uncensored model. The name also falsely implied lineage with Claude Opus. Before running any downloaded model: check the name for uncensored/ablated branding and fabricated lineage claims to closed frontier models. Prefer official `mlx-community` releases.

Model swap: Qwen3.6-27B-8bit → Qwen3.6-35B-A3B-8bit (2026-07-11)

Motivation: the dense 27B model was clearing ~16.8 tok/s through Codex, well under the ~30 tok/s target for a smooth interactive loop.

What was tried and ruled out:

- `antirez/ds4` ("DwarfStar") — a dedicated inference engine for DeepSeek V4 Flash (284B total / 13B active). Legitimate, MIT-licensed, real published benchmark of 34.27 tok/s / 25.9 tok/s (short/11.7K-context) on an M5 Max 128GB — the exact machine class here. Not adopted yet: beta-quality software, a separate engine/server from omlx (not a swap-in), and `ds4-agent` is explicitly alpha. Worth revisiting if the omlx-native option below stops being enough.
- GLM-5.2 and MiniMax-M3 top the open-weight intelligence leaderboards (Artificial Analysis) but are physically too large for 128GB (753B and 427B total params respectively, even at aggressive 4-bit quant).

What was adopted: `mlx-community/Qwen3.6-35B-A3B-8bit` — same family omlx already has custom kernels for (Qwen3.5/3.6), MoE with only ~3B active params per token, zero new integration (just `hf download` + `omlx restart`, no new alias structure, Codex/OpenCode stay wired exactly as before).

Measured results (own hardware, this M5 Max 128GB):

- Raw `mlx_lm.benchmark`, short prompt: **~104 tok/s** generation (vs ~16.8 tok/s baseline)
- Raw `mlx_lm.benchmark`, ~11.7K context: **~94 tok/s** generation — MoE barely degrades under long context, unlike the dense model
- Through the live omlx server (`/v1/chat/completions`): ~88 tok/s
- Real `codex exec` run, identical prompt on both models: 42.3s wall-clock (27B, prompt-cache warm) vs 14.3s (35B-A3B, prompt-cache cold — a harder case for it) while producing 53% *more* output tokens
- **Net: ~5.6x faster generation throughput, ~2.9x faster wall-clock on a real Codex task**

Quality tradeoff (Qwen's own published benchmarks, direct comparison of these two exact checkpoints):

Benchmark	Qwen3.6-27B (old default)	Qwen3.6-35B-A3B (new default)	Gap
SWE-bench Verified	77.2	73.4	-3.8
SWE-bench Pro	53.5	49.5	-4.0

Benchmark	Qwen3.6-27B (old default)	Qwen3.6-35B-A3B (new default)	Gap
Terminal-Bench 2.0	59.3	51.5	-7.8
LiveCodeBench v6	83.9	80.4	-3.5
MMLU-Pro	86.2	85.2	-1.0
GPQA Diamond	87.8	86.0	-1.8
AIME26	94.1	92.7	-1.4

The dense model wins every benchmark, but mostly by 1-4 points; Terminal-Bench (closest proxy to real agentic CLI tool-use) shows the largest gap at ~8 points. Judged worth it for the ~3-5.6x speed gain in an interactive loop where iteration speed compounds.

Fallback: the old 27B weights are kept on disk (not deleted). For the rare task where you want max quality over speed:

```
omlx launch codex --model mlx-community--Qwen3.6-27B-8bit # DELETED 2026-09-17 - do not run, see $5a
```

No non-abliterated fallback is installed. For max quality over speed, ou → omlx launch opencode --model Qwen3.8-27B-Uncensored-8bit (~18 tok/s), noting it is abliterated.

Model swap: Qwen3.6-35B-A3B-8bit → Qwen3-Coder-Next-8bit (2026-07-12, Codex only)

Motivation: newer Qwen3-Next hybrid architecture, still only ~3B active params (so generation speed stays close to the 35B-A3B default) but 80B total params and 256K native context vs the old model's much smaller window — worth it given 128GB unified memory has plenty of headroom for the 8bit quant (~79GB on disk).

Adopted: mlx-community/Qwen3-Coder-Next-8bit via omlx's bundled hf CLI (/opt/homebrew/Cellar/omlx/<version>/libexec/bin/hf download mlx-community/Qwen3-Coder-Next-8bit), same quant precision (8bit) as both prior defaults for consistency. ~/.codex/config.toml updated: model = "mlx-community--Qwen3-Coder-Next-8bit" (provider stays omlx).

Verified: started the omlx server, confirmed the model in GET /v1/models , sent a direct /v1/chat/completions test request (loaded + responded correctly, ~12.4s cold load), then ran a real codex exec end-to-end (quicksort/mergesort/heapsort request) and got correct, well-formed output.

Measured results (own hardware, this M5 Max 128GB), same methodology as the earlier swap:

	Qwen3.6-27B-8bit (dense)	Qwen3.6-35B-A3B-8bit (MoE)	Qwen3-Coder-Next-8bit (MoE)
Short-context generation	~16.8 tok/s	~104 tok/s raw / ~88 tok/s server	~79 tok/s raw (mlx_lm.benchmark -p 64 -g 256)
~11.7K-context generation	not measured	~94 tok/s	~74 tok/s raw (mlx_lm.benchmark -p 11700 -g 256)
Peak memory	27GB-class	~37-39GB	~85-87GB
Real codex exec wall-clock (identical quicksort/mergesort/heapsort prompt)	42.3s	14.3s (cold)	9.0s (clean re-run — see caveat below)

Takeaways: Coder-Next is slower per-token than 35B-A3B raw (~74-79 tok/s vs ~88-104 tok/s) — expected, it's a much bigger model (80B vs 35B total) even at the same ~3B active params, so routing/attention overhead is higher. It still shares 35B-A3B's key property: barely slows down under long context (79→74 tok/s, ~6% drop) vs. how much a dense model degrades. The 9.0s codex exec result being the fastest of the three shouldn't be over-read as "Coder-Next is raw-faster" — it's one sample, likely favorable output length plus a warm server, not a contradiction of the raw benchmark numbers above.

Caveat — don't benchmark back-to-back with other heavy loads: the first codex exec attempt this session measured 4:12 (vs the clean 9.0s above). Cause: two mlx_lm.benchmark runs (each memory-mapping ~85-87GB of weights) had just run immediately before it, evicting Coder-Next's pages from the page cache and forcing a slow cold reload under memory contention when the server needed them. Not a model or config problem — an artifact of testing methodology. Let large local-model loads settle before timing the next one.

Only `co` /Codex was switched — `oc` /OpenCode's model is hardcoded in the `home.nix` alias (not read from `config.toml`), so it's still on `Qwen3.6-35B-A3B-8bit` until deliberately changed. (*Superseded 2026-08-17: `oc` was swapped to `Qwen3-Coder-Next-8bit` to match `co`.*)

Fallback: both older models (`Qwen3.6-35B-A3B-8bit`, `Qwen3.6-27B-8bit`) are kept on disk, same as always — swap the `model` line in `~/codex/config.toml` back if needed, no rebuild required (that file isn't Nix-managed):

No longer true as of 2026-09-17. Both Qwen3.6 fallbacks were **deleted** (63GB reclaimed), along with two never-completed stub downloads. Do not point `~/codex/config.toml` at `Qwen3.6-35B-A3B-8bit` or `Qwen3.6-27B-8bit` — those weights are gone and the server will fail to load them. There is currently **no** non-abliterated fallback installed; `ou` (`Qwen3.8-27B-Uncensored-8bit`) covers deep reasoning, and the next model downloaded is slated to become the new daily driver. See §5a.

§5a — Model lineup rework: two slots (2026-09-17)

Consolidated from five cached models (142GB) to two deliberate slots (~107GB). Deleted `Qwen3.6-27B-8bit` (28G), `Qwen3.6-35B-A3B-8bit` (35G), and two stub directories that had never finished downloading (`Qwen3.6-40B-Deckard-Heretic-...`, `Qwen3.6-27B-nvfp4`). Net 63GB reclaimed.

Alias	Model	Size	Role
<code>oc</code>	<code>mlx-community--Qwen3-Coder-Next-8bit</code>	79GB	Daily driver: agentic coding (unchanged)
<code>ou</code>	<code>Qwen3.8-27B-Uncensored-8bit</code>	27.5GB	Deep reasoning + refusal bypass

A third slot was specced and deliberately deferred. `mlx-community/Qwen3.8-27B-8bit` (29.5GB) was going to be the max-quality non-abliterated reference, replacing the deleted Qwen3.6-27B in that role. It was dropped on 2026-09-17 in favour of waiting: the *next* model downloaded becomes the new daily driver instead. Two consequences worth knowing:

- There is **no non-abliterated counterpart** on this machine. `ou` was meant to be A/B'd against the clean base build of the same model — both derive from `Qwen/Qwen3.8-27B`, so comparing them would have isolated ablation from model choice. Without it, odd behaviour from `ou` is unattributed: could be ablation damage, could just be the model.
- `oc` remains the daily driver until that next download lands.

Why a 27B at all, rather than something larger. On 128GB every bigger model forces a 2-bit quant, which costs more capability than the extra parameters buy back. GLM-5.3 (max) needs 217GB for its smallest build; GLM-5.3-Flash fits only at `2bit-lite` (+141% perplexity, 77.19% top-1 agreement); Qwen3.8-Flash-Next is 163GB in MLX 4-bit; Kimi K3's smallest real quant is 567GB; MiniMax M3 needs ~118GB at IQ2_XXS. Qwen3.8's open lineup has no mid-tier — 27B dense, then 180B Flash-Next, then 2.4T-A95B. Qwen3.8-27B scores 52 on the Artificial Analysis Intelligence Index, tying DeepSeek V4 Flash at roughly a tenth of the memory and taking no quantization damage to get there.

Why not DeepSeek V4 Flash. Rejected twice over. First on merit: the 128GB-viable build is ~91GB at Q2 *and* ablated — two stacked degradations hitting exactly the reasoning and long-context robustness that justified reaching for it. Then on hard feasibility: `omlx` bundles `mlx_lm`, whose `models/` directory ships `deepseek.py`, `deepseek_v2.py`, `deepseek_v3.py`, `deepseek_v32.py` and **no v4** — upstream `mlx-lm` [issue #1281](#) is open, because V4 introduces Compressed Sparse Attention and Heavily Compressed Attention that need new implementation work. The `mlx-community/DeepSeek-V4-Flash-*` quants on HuggingFace are publishable files with no runtime on this stack. (`dflash_mlx` in site-packages is *DFlash speculative decoding*, not DeepSeek Flash — easy to misread.) Running V4 Flash at all requires antirez's `ds4`, a separate non-MLX engine outside `omlx` entirely, whose `ds4-server` also defaults to `127.0.0.1:8000` and would collide with `omlx`.

No upgrade exists for `oc` either. There is no Qwen3.8-Coder; Qwen3-Coder-Next (Feb 2026) is still the current coder model, and the newer Qwen3.8-Flash-Next is a general model that doesn't fit at 163GB in MLX.

Model id resolution differs between the two aliases — worth remembering when adding the next one:

- `oc` uses `mlx-community--Qwen3-Coder-Next-8bit`: resolved from the HF cache, repo id with `/` → `--`.
- `ou` uses `Qwen3.8-27B-Uncensored-8bit`: a **directory name** under `~/omlx/models`. `omlx serve` discovers each subdirectory containing `config.json` + `*.safetensors` and names the model after the folder. The orcarouter repo ships quants in `8-bit/` etc. subfolders, so the download must be flattened into a single model directory or `omlx` won't see a valid model. Also note `omlx` only rescans at startup — after adding weights, `omlx restart` or the server keeps serving its old list (including models you have since deleted).

`hf` (`python3Packages.huggingface-hub`) is now declared in `home.nix` rather than relying on `omlx`'s bundled copy at `/opt/homebrew/Cellar/omlx/<version>/libexec/bin/hf`.

Both aliases launch OpenCode, not Codex — including `ou`, even though the Qwen3.6-27B fallback that once filled a similar role used `omlx launch codex`. Reason: the §5 guidance below — prefer OpenCode for context-heavy work, since Codex fabricates its context-window metadata. Deep reasoning is the context-heavy case.

Measured throughput (2026-09-17, M5 Max 128GB)

Run `./docs/bench-models.sh` to reproduce. Numbers are end-to-end (output tokens / wall clock), cold MLX load excluded via a warmup request:

Model	short prompt	~3k prompt	bandwidth ceiling
Qwen3-Coder-Next-8bit (MoE, ~3B active)	78.1 tok/s	71.8 tok/s	~186 tok/s
Qwen3.8-27B-Uncensored-8bit (dense, 27B active)	18.0 tok/s	17.1 tok/s	~20.8 tok/s

Coder-Next is ~4.3x faster. It is 2.7x larger on disk (79GB vs 29.5GB) but activates only ~3B params per token, while the dense 27B reads all 27B. Generation speed tracks active params against memory bandwidth, not total size. The dense figure sits just under its 20.8 tok/s ceiling (29.5GB ÷ 614GB/s), confirming it is bandwidth-bound — and it lines up with the ~16.8 tok/s the old dense Qwen3.6-27B measured in §5. Both models pass the reply-exactly-OK diagnostic, so neither will reproduce the "produced no final answer" failure.

Two measurement gotchas the script now guards against, both discovered building it:

- 1. **Short generations are unmeasurable.** omlx packs several tokens into one SSE chunk. At ~30 output tokens the decode span is mostly buffering jitter, which produced a nonsensical 171 tok/s for a model whose hard ceiling is 20.8. Tests now force 512-token outputs, and any decode figure above the bandwidth ceiling is flagged as an artifact rather than printed as fact.
- 2. **The VLM serving path does not stream incrementally.** Qwen3.8-27B is a vision model, and omlx returns its entire response in a single SSE chunk — TTFT equals wall clock. Prefill and decode cannot be separated there at all, so the script reports `n/a` and says why. Coder-Next streams normally. This is why end-to-end tok/s is the headline metric: it is the only one comparable across both paths.

Codex's fabricated context-window metadata (investigated, not fixed — by design)

Codex's TUI reports "Model metadata... not found" and internally uses `model_context_window: 258400` for any locally-served model — not from omlx, which correctly reports `max_model_len: 65536` via its own `/v1/models` endpoint. Codex's bundled model catalog just doesn't recognize custom/local models and substitutes a generic, provider-agnostic fallback (258400 = 272000 × 0.95), confirmed identical for GPT-5.5 on an unrelated provider in an upstream GitHub issue. Confirmed again 2026-07-12 via `codex debug models`: the entire bundled catalog is only 8 entries, all OpenAI's own hosted models (`gpt-5.5`, `gpt-5.6-sol`, etc.) — no locally-served model can ever match it, so this warning is permanent and provider-agnostic, not specific to whichever local model is currently the default.

Deliberately **not** patched via Codex's `model_context_window` config override: a confirmed upstream bug (openai/codex #16068) means manually setting that value can permanently break auto-compaction after the first context overflow. Safe mitigations instead: prefer `oc` for anything context-heavy (already accurate), and for long `co` sessions run `/compact` manually rather than trusting Codex's automatic trigger. (Almost reintroduced this override during the 2026-07-12 model swap — caught and reverted before it landed; worth re-checking `~/.codex/config.toml` doesn't have a stray `model_context_window` line after any future swap.)

GPU memory & context-window hardening (2026-07-10)

omlx hard-crashed once under memory pressure. Root cause: macOS resets the `iogpu.wired_limit_mb` sysctl to a low default on every reboot, undercutting omlx's own internal ~122GB target, so instead of gracefully rejecting an oversized request it ran out of wired memory and hard-panicked. Fixed with two durable changes:

- `iogpu.wired_limit_mb=124928` (122GB) via a `launchd.daemons` entry in `configuration.nix` with `RunAtLoad = true` — not `system.activationScripts`, which has a known nix-darwin issue where it doesn't reliably re-run after reboot (only on `darwin-rebuild switch`)
- `max_context_window_policy: 65536` in `~/.omlx/settings.json` (under `sampling`) — caps requests server-side so an oversized context is rejected up front instead of exhausting memory

Adding another local model later

```
/opt/homebrew/Cellar/omlx/<version>/libexec/bin/hf download mlx-community/<repo>
omlx restart # required – server doesn't hot-reload; this triggers a rescan
```

Benchmark before switching defaults:

```
/opt/homebrew/Cellar/omlx/<version>/libexec/bin/mlx_lm.benchmark --model <repo> -p 64 -g 256 -n 3
/opt/homebrew/Cellar/omlx/<version>/libexec/bin/mlx_lm.benchmark --model <repo> -p 11700 -g 256 -n 2
```

6. Voice Input — OpenSuperWhisper + Parakeet v3

Local, free, open-source dictation. Installed as a cask, declared in `configuration.nix`.

Model	Why
Whisper V3 Large/Medium/Small	not used — slower, hallucinates during silence
Parakeet v2	marginally more accurate on English only (6.05% vs 6.32% WER), no multilingual
Parakeet v3 — chosen	beats Whisper Large V3 on accuracy (6.32% vs 7.44% WER) at ~23x the speed, zero silence hallucination, adds 25-language support at negligible cost

Known limitation: vocabulary/hotword priming (feeding it jargon like "herdr", "omlx") only works with Whisper models in this app, not Parakeet. Accepted tradeoff — fall back to typing or manual correction on jargon-heavy prompts.

7. Not Yet Built — Kokoro TTS (Voice Output)

omlx already bundles `mlx-audio`, which includes the Kokoro TTS model — genuinely on the machine right now, just unused. Neither Claude Code nor Codex has a built-in "read the response aloud" feature. Building it would take a small hook/script: capture an agent's final response text → POST to omlx's TTS endpoint → play the resulting audio. Candidate integration points: a Claude Code Stop hook, or a wrapper skill. Not started, flagged so it isn't forgotten.

8. Why This Is a Different Way of Working

Every point of friction in the loop has a purpose-built fix, and they compose:

- **Typing** → voice input at ~3x typing speed, global and agent-agnostic
- **Reading walls of text** → lavish turns a plan into an annotatable visual artifact
- **Reviewing every diff** → no-mistakes catches what a human reviewer would, with recorded evidence
- **One agent at a time** → treehouse makes parallel worktrees free
- **Managing multiple sessions** → firstmate is a single point of contact for a crew
- **Tool inefficiency** → gh-axi/chrome-devtools-axi cut token cost and latency on GitHub/browser work
- **Cost/availability** → omlx gives Codex and OpenCode a zero-marginal-cost, fully local, fully private backend, with Claude Code still available for the highest-stakes work

None of this replaces judgment — it moves the job from producing and checking individual changes to directing a process that produces and checks itself.

A real day: morning voice-driven planning in lavish → midday parallel scaling across treehouse worktrees → afternoon judgment proportional to risk, not diff size → overnight autonomous gnhf loop → anytime cost-free iteration via `co / oc`, reaching for `cc` when stakes call for Claude's ceiling.

9. Maintenance, Upgrading & Future-Proofing

Golden rule: everything durable is declared in `~/github/JalenMickey/dotfiles/configuration.nix` (Homebrew taps/brews/casks) or `home.nix` (nixpkgs packages, env vars, session PATH). `homebrew.onActivation.cleanup = "zap"` is intentional and will silently remove anything installed ad-hoc on the next `./rebuild.sh`. Declare before your next rebuild or expect it gone.

Never run `brew install/reinstall/upgrade` at the same time as `./rebuild.sh`. Both touch the Homebrew Cellar and will lock-collide. Wait for one to finish before starting the other.

`sudo` commands (including `./rebuild.sh` itself) fail when run non-interactively — always run these yourself in a real terminal, not via an agent's background shell.

Known gotchas (all hit and resolved across setup sessions)

Symptom	Real cause	Fix
<code>home.nix</code> secret-file pattern (<code>omlx-api-key.local</code> etc.) silently resolves to <code>""</code> after a rebuild, even though the file exists and is readable	<code>darwin-rebuild switch --flake</code> runs Nix evaluation in pure mode by default; <code>builtins.pathExists / readFile</code> on any absolute path outside the flake's own store copy silently returns <code>false</code> /errors instead of reading the real file — confirmed directly with <code>nix eval --impure</code> (true) vs <code>nix eval</code> (false) against the same path	<code>rebuild.sh</code> now passes <code>--impure</code> to <code>darwin-rebuild switch</code> . Required for this secret-file pattern to ever work — re-check this flag survives if <code>rebuild.sh</code> is ever rewritten
New <code>home.nix</code> env var doesn't show up anywhere, even a fresh terminal or fully quitting and reopening WezTerm	<code>herdr</code> is a persistent-session multiplexer (<code>herdr</code> launches by "attach to persistent session", not a fresh one) — quitting/reopening the terminal app just reattaches the same long-lived pane process with its original environment. <code>herdr server stop</code> doesn't kill that pane either. Even <code>exec zsh -l</code> in that pane isn't enough by itself: <code>home-manager's</code> <code>hm-session-vars.sh</code> has its own re-source guard (<code>__HM_SESS_VARS_SOURCED</code>), and <code>exec</code> inherits environment variables from the replaced process — so the guard (already <code>1</code> from before the rebuild) blocks re-export of everything, including the fixed value	In the existing pane: <code>unset __HM_SESS_VARS_SOURCED __HM_ZSH_SESS_VARS_SOURCED; exec zsh -l</code> . Confirm with <code>echo \$OMLX_API_KEY</code> (or whatever var) before trusting it
App records/detects fine but silently does nothing (e.g. dictation doesn't paste)	macOS Accessibility/TCC grants don't apply to an already-running process	fully quit and relaunch the app after granting permission
<code>omlx</code> broken after a rebuild (bad interpreter path)	undeclared transitive dependency (<code>python@3.11</code>) got zap-removed as an "orphan"	declare every real dependency explicitly in <code>configuration.nix</code>
global <code>npm install -g</code> fails outright	this machine's npm prefix is a read-only Nix store path	use <code>npmx <pkg></code> for on-demand invocation
<code>omlx</code> hard-crashed under memory pressure	<code>iogpu.wired_limit_mb</code> resets to a low default every reboot	<code>launchd.daemons</code> <code>sysctl</code> entry + <code>max_context_window_policy: 65536</code> — see \$5
<code>co</code> breaks with unexpected argument <code>'--full-auto'</code>	Codex 0.128 removed <code>--full-auto</code>	use <code>--sandbox workspace-write --ask-for-approval never</code> — not <code>--dangerously-bypass-approvals-and-sandbox</code> , which removes the sandbox entirely
Codex reports fabricated context-window metadata (258400)	bundled catalog doesn't recognize local models, falls back to a generic constant	no safe config fix (upstream #16068); use <code>oc</code> for context-heavy work, <code>/compact</code> manually in long <code>co</code> sessions
<code>leetcode.nvim</code> errors "contains listed buffers" on <code>:Leet</code>	plugin refuses to launch in a session with other files already open, by design	fixed here via <code>plugins.non_standalone = true</code> in <code>leetcode.lua</code> — close with <code>:Leet exit</code>
<code>leetcode.nvim</code> sign-in: pasting the cookie does nothing	<code>Ctrl+Shift+V</code> / <code>Shift+Insert</code> / <code>"*p</code> don't reliably paste into the prompt in some terminals	try <code><C-r>*</code> (insert register), or your terminal's native paste; on Brave, strip the leading <code>cf_clearance=...</code> ; so the string starts at <code>csrftoken=</code>
<code>node</code> / <code>npm</code> abort with <code>Library not loaded: .../libada.3.dylib</code> (breaks <code>npmx gnghf</code> , and <code>opencode</code> / <code>oc</code> which is a Homebrew node dependent)	Homebrew <code>node</code> links <code>libada.3</code> , but its <code>ada-url</code> dependency auto-upgraded to 4.0.0 (only <code>libada.4</code> exists) and node wasn't relinked - Homebrew <code>onActivation.autoUpdate = true</code> is the trigger	<code>brew reinstall node</code> pulls a current node bottle linked to <code>libada.4</code> . Do NOT <code>brew uninstall node</code> : <code>opencode</code> depends on it. Note this node is undeclared cruft - you declare <code>nixpkgs nodejs_22</code> (works at v22, shadowed on PATH), so a rebuild would zap it anyway

Symptom	Real cause	Fix
<code>brew reinstall/install</code> aborts with <code>Error: unknown install step: remove</code>	nix-homebrew's patched brew (6.0.1, <code>zhaofengli/nix-homebrew</code>) is behind homebrew-core; the current formula uses a post-install step the patched brew doesn't implement	the bottle itself usually still pours and links (node came out usable), so verify with <code>node -v</code> before panicking. Durable fix: <code>nix flake update nix-homebrew && ./rebuild.sh</code> - never mid-brew
<code>gnhf</code> (or any <code>oc /OpenCode agentic run</code>) burns iterations with "OpenCode produced no final answer", 0 commits, then bails after 3 strikes - even though the omlx server is up and answers <code>curl</code>	the model was pointed at a reasoning model (<code>Qwen3.6-35B-A3B</code>), which spends its whole output budget in a <code><think></code> stream and never lands a clean tool call. Proof: <code>curl</code> the model with "reply exactly OK" and it returns "Here's a thinking process:\n\n1." with <code>finish_reason: length</code> instead of <code>OK / stop</code> . Reasoning models are fine for chat, wrong for agentic tool-calling loops	point OpenCode at a coder model instead - <code>~/.config/opencode/opencode.json</code> <code>"model": "omlx/mlx-community--Qwen3-Coder-Next-8bit"</code> (declare it under <code>provider.omlx.models</code> too). Verify: <code>curl</code> it with "reply exactly OK" - a coder model returns <code>OK / finish_reason: stop</code> . Same rule for Codex/ <code>co</code> , which already defaults to the Coder model (see §5). First call per model is a ~1min cold MLX load, not a hang

Update commands (verified)

What	Command
Skills (lavish, axi family, skill-creator)	<code>npk skills update -g</code>
Codex CLI	<code>codex update</code>
OpenCode	<code>brew upgrade opencode</code> (declared in <code>configuration.nix</code> , so <code>./rebuild.sh</code> also keeps it current)
herdr	<code>herdr update</code> (or <code>herdr channel set stable preview</code>)
firstmate	<code>/updatefirstmate</code>
gnhf	nothing to do — <code>npk gnhf</code> always resolves latest
treehouse / no-mistakes	re-run the original install script (idempotent)
omlx	<code>brew upgrade jundot/omlx/omlx</code> — recompiles from source (~3 min); never alongside <code>./rebuild.sh</code>
Copilot CLI	<code>copilot update</code> — the cask is tagged <code>auto_updates</code> , so <code>brew upgrade</code> won't manage it; the app updates itself
gh, node, everything else in <code>home.packages</code>	<code>nix flake update</code> then <code>./rebuild.sh</code>

Periodic health checks

- `gh auth status` — tokens can expire
- `copilot login` (re-run if `gc` starts prompting to authenticate — Copilot tokens can also expire)
- `omlx diagnose menubar` — built-in self-diagnostic
- Check disk usage under `~/.omlx/models` and `~/.cache/huggingface` — local models are 20-80GB+ each and accumulate
- Before downloading any new local model: re-run the uncensored/abliterated-branding sanity check from §5
- Re-run `./rebuild.sh` occasionally even with no known changes — catches config drift early

Secrets hygiene

The dotfiles repo is public. Pattern used for `OMLX_API_KEY` — reusable for any future credential:

```
home/<name>.local          # plaintext value, gitignored (home/*.local)
home.nix reads it via:
  lib.optionalString (builtins.pathExists file) (lib.strings.trim (builtins.readFile file))
```

This keeps the *shape* of the config declarative and public while the actual secret value never enters git history.

This pattern requires `--impure`. `builtins.pathExists` / `readFile` on the secret file's absolute path silently fails under Nix's default pure evaluation for flakes — see the gotchas table above. `rebuild.sh` already has this flag; don't drop it when adding the next secret.